AN EXTENSION OF ADEQUATE FUNCTORS IS ADEQUATE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Kernels and cokernels of maps of adequate functors are adequate. Same chapter, Lemmas lemma-kernel-adequate and lemma-cokernel-adequate, invoked at proof lines 920-923, 931-933 and 948-949. Let the successive representations of G as a kernel or cokernel be used.
- Every adequate functor is a quotient of a direct sum of linearly adequate functors. Same chapter, Lemma lemma-adequate-surjection-from-linear, invoked at proof lines 927-929. Produces the surjection L -> H used in the second reduction.
- An extension of a linearly adequate functor by an adequate one is adequate. Same chapter, Lemma lemma-extension-adequate-key, invoked at proof lines 935-937. Gives adequacy of each pullback G x_H L_i.
- Direct sums of adequate functors are adequate. Same chapter, Lemma lemma-colimit-adequate, invoked at proof lines 937-939. Gives adequacy of the direct sum of the pullbacks.

Source reference. Original lemma 06UX: sources/stacks-adequate/pinned-original-source.tex, source lines 468–468. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V2: sources/stacks-adequate/pinned-original-source.tex, source lines 684–684. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V4: sources/stacks-adequate/pinned-original-source.tex, source lines 734–734. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V6: sources/stacks-adequate/pinned-original-source.tex, source lines 781–781. Pointer outside the counted span, invoked once

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V3: sources/stacks-adequate/pinned-original-source.tex, source lines 709–709. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. Let $0 \to F \to G \to H \to 0$ be a short exact sequence of module-valued functors on Alg_A. If F and H are adequate, so is G.*

Proof. Choose an exact sequence $0 \to F \to \underline{M} \to \underline{N}$. If we can show that $(\underline{M} \oplus G)/F$ is adequate, then G is the kernel of the map of adequate functors $(\underline{M} \oplus G)/F \to \underline{N}$, hence adequate by Lemma 06V3. Thus we may assume $F = \underline{M}$.

We can choose a surjection $L \to H$ where L is a direct sum of linearly adequate functors, see Lemma 06UX. If we can show that the pullback $G \times_H L$ is adequate, then G is the cokernel of the map $\mathrm{Ker}(L \to H) \to G \times_H L$ hence adequate by Lemma 06V2. Thus we may assume that $H = \bigoplus L_i$ is a direct sum of linearly adequate functors. By Lemma 06V6 each of the pullbacks $G \times_H L_i$ is adequate. By Lemma 06V4 we see that $\bigoplus G \times_H L_i$ is adequate. Then G is the cokernel of

$$\bigoplus\nolimits_{i \neq i'} F \longrightarrow \bigoplus G \times_H L_i$$

where ξ in the summand (i, i') maps to $(0, \ldots, 0, \xi, 0, \ldots, 0, -\xi, 0, \ldots, 0)$ with nonzero entries in the summands i and i'. Thus G is adequate by Lemma 06V2. $\square$